Joint Communication and SAR Waveform Design Method via Time-Frequency Spectrum Shaping

Youshan Tan, *Student Member, IEEE*, Zhongyu Li, *Member, IEEE*, Jing Yang, Xianxiang Yu, Hongyang An, *Member, IEEE*, Junjie Wu, *Member, IEEE*, and Jianyu Yang, *Member, IEEE*

***Abstract*— Due to the division of the transceiver, a bistatic synthetic aperture radar (SAR) gains many advantages, such as forward-looking imaging and powerful anti-interference capabilities. In the meantime, information sharing (e.g., positions and status) between the transmitter and the receiver is required for SAR imaging. This article addresses the co-use waveform design for SAR-dual-functional radar communication (SAR-DFRC). To this end, we embed information into a time-frequency spectrum of the phase coded signal, which can be a feasible solution to SAR-DFRC, bringing the possibility of realizing a light-weighted, miniaturized, low-costed, spectrum reusable, and more confidential system. A novel time-frequency spectrum shaping (TFSS) SAR-DFRC architecture based on short-time Fourier transform (STFT) is proposed for the first time. The weighted peak sidelobe level (WPSL) is considered a figure of merit for SAR imaging performance. Information is embedded by nulling the time-frequency spectrum of the waveform. Here, we develop the majorization–minimization PSL-TFSS (MMPSL-TFSS) algorithm to solve the resulting nonconvex NP-hard optimization problem. The designed waveform can ensure imaging performance and obtain high communication capacity in the meantime. Experimental and numerical results highlight the effectiveness of the proposed SAR-DFRC framework for imaging performance and the secure transition to communication information.**

***Index Terms*— Alternating direction method of multipliers (ADMM), dual-functional radar communication (DFRC), short-time Fourier transform (STFT), synthetic aperture radar (SAR).**

I. Introduction

A SYNTHETIC aperture radar (SAR) has attracted much attention in civil and military fields because of its property of obtaining high-resolution microwave images and the capability of operating in the condition of all-day and all-weather, which makes SAR widely used in many remote sensing scenarios, such as earth mapping and disaster monitoring [1]. Due to the division of the transceiver, bistatic SAR obtains even more superiority [2]. The increased information content of scattering makes the feature extraction and classification easier [3], which can be worthwhile for the forest, vegetation, and soil resource management [4]. The forward-looking imaging capability makes it promising in military scenarios [5], [6], [7]. Some successful experiments and applications have been carried out, such as the bistatic side-looking SAR experiment [8], bistaic forward-looking SAR experiment [9], the bistatic SAR-ground moving target indication (SAR-GMTI) experiment [10], and the satellite-airborne bistatic SAR experiment [11].

While in the bistatic SAR systems, information (e.g., position, velocity, track, status, beam, and synchronization [12], [13]) sharing between the transmitter and the receiver is indispensable for bistatic SAR imaging processing, where the radio communication link is a common way [14].

The radio communication link plays a vital role in the existing bistatic SAR system. It is also an essential factor affecting the imaging quality of bistatic SAR [15]. However, establishing the radio communication link requires additional communication units, which will not only increase the cost but also be unfavorable to the miniaturization and lightweight of the system and will occupy a certain amount of additional spectrum resources [16]. Meanwhile, it is likely to be vulnerable to interferences and interceptions, which will result in severe consequences in military scenarios.

Joint radar-communications (JRCs) technology may be an effective way to solve the above problems. In the field of SAR imaging, there has been some research related to joint SAR communication (JSC). Experiments were carried out to validate a feasible method to perform data transmission and SAR imaging via the multiple-input multiple-output-SAR (MIMO-SAR) system [17]. A cyclic prefix (CP)-based orthogonal frequency-division multiplexing (OFDM) SAR imaging system to gain high range resolution was proposed in [18]. However, as it is mentioned in this article, the OFDM usually comes up with a high peak-to-average power ratio (PAPR). A JRCs receiver was designed to realize JSC [19]. Throughout the existing related studies, communication and SAR are still two sets of hardware in most JSC, which is not conducive to miniaturization. OFDM SAR has the problem of high PAPR, which is not conducive to practical applications. By comparison, dual-functional radar communication (DFRC) technology has natural advantages in high spectral efficiency, low cost, miniaturization, and lightweight as a joint system

Manuscript received 20 July 2022; revised 9 October 2022; accepted 7 December 2022. Date of publication 26 December 2022; date of current version 30 December 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62171084, Grant 62101096, Grant 62101097, Grant 62001423, Grant 61901088, Grant 61922023, Grant U19B2017, and Grant 61871080; in part by the ChangJiang Scholar Program, in part by the 111 Project under Grant B17008; in part by the China Postdoctoral Science Foundation under Grant 2020M680147 and Grant 2021T140096; in part by the Open Fund of Science and Technology on Electronic Information Control Laboratory; and in part by the Postdoctoral Innovation Talent Support Program under Grant BX2021058. *(Corresponding author: Zhongyu Li.)*

Youshan Tan, Zhongyu Li, Xianxiang Yu, Hongyang An, Junjie Wu, and Jianyu Yang are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: zhongyu_li@hotmail.com).

Jing Yang is with the School of Electrical and Information Engineering, Zhengzhou University, Zhengzhou 450001, China (e-mail: yangjinguestc@163.com).

Digital Object Identifier 10.1109/TGRS.2022.3230439

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with a shared, single transmitted waveform [20]. As a result, the DFRC waveform is more suitable for SAR systems to realize information sharing compared with other integration technologies.

DFRC waveforms have been extensively studied in the past decade, covering many aspects. The related research can be mainly divided into two categories [21]: communication signal sensing and radar signal embedding information. The former studies cover many communication waveforms. Zhou et al. [22] decreased the peak sidelobe level (PSL) by offsetting the binary phase shift keying (BPSK) and quadrature phase shift keying (QPSK) sequence phase to ensure the radar and communication performance. The PAPR constrained OFDM signal was optimized by pareto-optimal solution [23]. Hassanien et al. [24] embedded the phase shift keying (PSK) code into the phase differences between orthogonal signals. Information was modulated by minimum shift keying (MSK), where the linear frequency modulation (LFM) signal is used as the carrier [25]. The M-sequence was mixed into the phase of LFM for integration [26]. Information was embedded into LFM subcarriers with different frequency modulation rates through fractional Fourier transform (FRFT) [27]. Nevertheless, these LFM-based waveforms usually cannot carry much information. Other research mainly modulates information in the domain of time and frequency [28], [29], space [30], space frequency [31], code [20], and index [32] domain of a signal. Moreover, [21] skillfully embeds information in the ambiguity function [33].

Although some mentioned DFRC waveforms can achieve a high data rate and have good PSL performance, confidentiality is also an important issue that must be carefully considered [34], [35]. Since it will result in severe consequences if the shared information (e.g., coordinate, azimuth, and altitude) of the bistatic SAR is decoded, higher demands are placed on confidentiality to the SAR-DFRC system. Nevertheless, most DFRC waveforms can only encrypt the information through coding. From this point of view, the confidentiality of these DFRC waveforms has no apparent advantages over the radio communication link. In other words, it is still vulnerable to interceptions.

To realize the JSC with DFRC waveform while ensuring confidentiality, we propose a novel DFRC waveform design method via time-frequency spectrum shaping (TFSS), since time-frequency transform naturally requires the transform window length and overlap length to be known. Namely, the modulation procedure is encrypted by the time-frequency transform, providing our waveform the encryption capability that most DFRC waveforms do not have. For demodulating convenience, the short-time Fourier transform (STFT) is considered the time-frequency transform in this article.

This article proposes a DFRC waveform design method via STFT spectrum shaping for SAR-DFRC. Weighted PSL (WPSL) of the autocorrelation function (ACF) is used as the objective function to ensure SAR imaging performance, and STFT spectrum nulling is used as the constraint to embed information. The waveform is optimized through the ADMM framework. The contributions of this work are highlighted as follows.

1) *Novel SAR-DFRC Architecture:* We propose a novel SAR-DFRC architecture by suppressing WPSL of the ACF to maintain SAR imaging performance in the range direction and nulling the STFT spectrum to embed the transmitter's information (e.g., position, velocity, altitude, track, status, beam, radar parameters, and synchronization information). We divide the STFT spectrum into two areas: passband and stopband. The passband is used for embedding information, which is constrained to be as low as possible. The rest of the passband of the STFT spectrum is constrained to be as high as possible to reduce the bit error rate (BER). The stopband is used for suppressing the energy leakage out of the bandwidth of the designed phase coded signal. Then, dual-function SAR imaging and communication can be realized.
2) *Establishing the Model of TFSS:* We establish and derive the general expression of the time-frequency cell, of which the whole STFT spectrum consists, along with the MMPSL-TFSS algorithm to solve the shaped waveform. The final expression indicates that any time-frequency cell of STFT can be formulated as a transform vector that multiplies the signal vector. The proposed method can bring three major benefits: For communication capacity, the time-frequency domain has one more dimension than the mere time or frequency domain, allowing more information to be embedded. For SAR imaging performance, the higher degree of freedom in the time-frequency domain compared with the mere frequency domain allows for easier sidelobe control. For confidentiality, modulation encryption is an exclusive capability of the time-frequency transform.
3) *Experimental and Numerical Validation:* We apply the experimentally generated waveforms to simulated bistatic SAR imaging whose scattering coefficients are obtained by an actual SAR image. The result shows that our measured waveforms can maintain the imaging performance compared with the LFM signal with the same bandwidth. Information is well encrypted in the time-frequency domain. Meanwhile, our waveform can achieve a low BER performance.

The symbol description is given in Table I. The rest of this article is organized as follows. In Section II, the bistatic SAR geometry and the proposed SAR-DFRC architecture are given. In Section III, the problem is formulated, and the algorithm to solve the problem is introduced. The experimental and numerical results and their analysis are given in Section IV. Conclusions are presented in Section V.

II. System Model

In this section, the bistatic SAR geometry and the proposed SAR-DFRC architecture are illustrated.

A. Bistatic SAR Geometry

Fig. 1 shows the spotlight bistatic SAR geometry. With the integrated waveform transmitted, the traditional communication link can be replaced by the SAR-DFRC system.

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TABLE I
NOTATIONS

- Bold small letter, e.g., $\mathbf{f}$ denotes vector. Bold capital letter, e.g., $\mathbf{F}$ denotes matrix. $f(i)$ denotes the i th element of $\mathbf{f}$. $\mathbf{F}(i,j)$ denotes the (i,j) th element of $\mathbf{F}$.
- $(\cdot)^*$, $(\cdot)^T$ and $(\cdot)^H$ denote the conjugate, the transpose and conjugate transpose, respectively.
- $\|\mathbf{x}\|$ denotes the Euclidean norm of $\mathbf{x}$.
- $\mathbb{C}^{N\times M}$, $\mathbb{R}^{N\times 1}$, $\mathbb{C}^{N\times 1}$ respectively denote the sets of $N\times M$-dimensional matrices of complex, $N\times 1$-dimensional vectors of real numbers and $N\times 1$-dimensional vectors of complex numbers.
- Letter j represents imaginary unit (i.e., $j=\sqrt{-1}$).
- $\Re\{\cdot\}$ and $\arg(\cdot)$, respectively, represent the real part and argument of a complex valued scalar, vector or matrix.
- $|x|$ represents the modulus of x.
- $\mathbf{Diag}(\cdot)$ indicates the diagonal matrix formed by the components of vector argument.
- $\lfloor\cdot\rfloor$ is the floor function.
- $Card\{\cdot\}$ denotes the number of elements of the set.
- $[a:\Delta:b]$ means the range $[a,b]$ with spacing Δ.

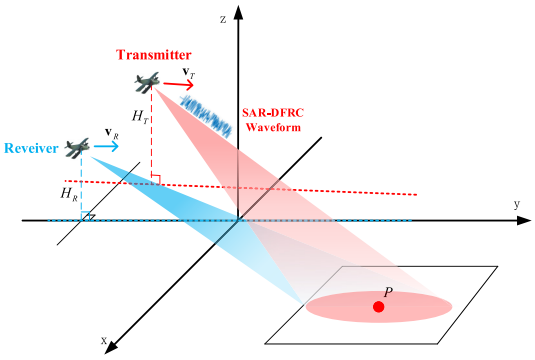


Fig. 1. Bistatic SAR geometry configuration.

Initial coordinates of the corresponding transmitter and receiver are, respectively, expressed as $\mathbf{p}_T=[X_T,Y_T,H_T]$ and $\mathbf{p}_R=[0,Y_R,H_R]$. The velocities of the two platforms above are $\mathbf{v}_T$ and $\mathbf{v}_R$, respectively. Without loss of generality, the receiver is assumed to move along the positive direction of the y-axis, and the transmitter is assumed to move along a straight line. Then, $\mathbf{v}_T$ and $\mathbf{v}_R$ can be decomposed as $\mathbf{v}_T=[v_{Tx},v_{Ty},v_{Tz}]$ and $\mathbf{v}_R=[0,v_{Ry},0]$. For a scattering point P in the XOY plane with coordinate $(x_p,y_p,0)$, according to the stop and go hypothesis, the instantaneous bistatic range $R_s(\eta)$ to the point P can be formulated as follows:

$$\begin{aligned} R_s(\eta) &= R_T(\eta)+R_R(\eta) \\ &= \|\tilde{\mathbf{p}}_T+\mathbf{v}_T\eta\|+\|\tilde{\mathbf{p}}_R+\mathbf{v}_R\eta\| \end{aligned} \tag{1}$$

where

$$\begin{aligned} \tilde{\mathbf{p}}_T &= \left[X_T-x_p, Y_T-y_p, H_T\right] \\ \tilde{\mathbf{p}}_R &= \left[x_p, Y_R-y_p, H_R\right]. \end{aligned}$$

B. Proposed SAR-DFRC Architecture

Diagram of the proposed SAR-DFRC system architecture is shown in Fig. 2. Therein, information is embedded into the STFT spectrum of a phase coded signal.

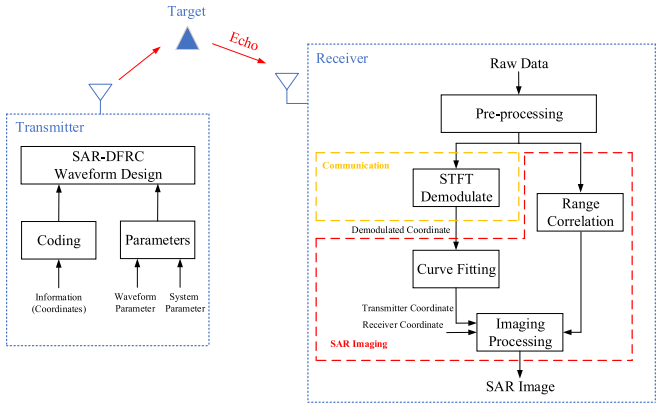


Fig. 2. Diagram of the SAR-DFRC architecture.

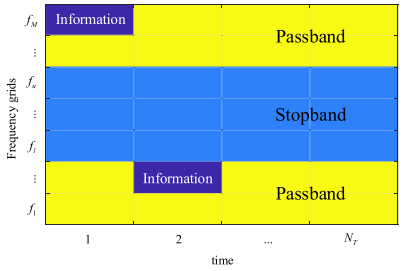


Fig. 3. Schematic of information embedding in time-frequency domain.

A schematic of the information embedding is shown in Fig. 3. For better BER performance, the normalized frequency points in the STFT spectrum are divided into a total of M frequency grids with m normalized frequency points in each grid. Frequency grids $[f_l,f_u]$ are considered as stopband, preventing the energy leakage out of bandwidth. Specifically, the remaining grids $[f_1,f_{l-1}]$ and $[f_{u+1},f_M]$ can be set to zero to embed information.

Therefore, the theoretical communication capacity transmitted in one pulse can be formulated as follows:

$$E=\left\lfloor T\log_2\sum_{i=0}^{\hat{L}}C_M^i\right\rfloor \text{ (bit)} \tag{2}$$

where $\hat{L}$ is the maximum number of simultaneous null frequency grids in one time grid. The maximal number of the time grids in the STFT spectrum can be formulated as follows:

$$N_T=\left\lfloor\frac{N-O}{L-O}\right\rfloor \tag{3}$$

where O is the overlap length, and L is the length of STFT window function. Since the time-frequency transform requires L and O to be known, the two parameters are preloaded to the transmitter and the receiver, which increases the difficulty of demodulation for eavesdroppers. The achievable data rate in bits per second (bps) can be expressed as follows:

$$R=E\times \text{PRF} \tag{4}$$

where PRF is the pulse repetition frequency.

The coding procedure is achieved by mapping all the statuses to the quantified spatial coordinates. Different positions

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of multiple null frequency grids in the STFT spectrum form a mass of symbols. In this manner, an alphabet with mutual mapping of the STFT spectrum and spatial coordinates is established. Meanwhile, the WPSL of the ACF is suppressed to ensure SAR imaging performance.

Let $x(t) = e^{j\varphi(t)}$ denote the transmitting signal. Then, the received baseband signal of scattering point P can be expressed as follows:

$$S(t,\eta) = \sigma_m \omega_r\left(t - \frac{R_s(\eta)}{c}\right) \times \exp\left(-j2\pi f_c \frac{R_s(\eta)}{c}\right) \times \exp\left(j\varphi\left(t - \frac{R_s(\eta)}{c}\right)\right) \quad (5)$$

where ω_r represents the range envelope of the signal after pulse compression, σ_m is the complex scattering coefficient of P, f_c is the carrier frequency, c represents the light speed, t represents the fast time, and η represents the azimuth slow time.

As shown in the receiver block of Fig. 2, after receiving and preprocessing the echo, raw data are parallel processed for communication and SAR imaging, respectively. The STFT is performed with transform window and overlap length to demodulate the embedded information. The demodulation process is realized by the threshold decision. By comparing with the threshold, the frequency grids shown in Fig. 3 are mapped to a binary plane. Then, the transmitted coordinates can be decoded. Simultaneously, the echo is compressed in the range direction. Next, the demodulated coordinates are curve fit to calculate the remaining transmitter coordinates corresponding to the reflected echo in each pulse. Finally, the SAR data after range correlation can be focused through imaging processing.

The proposed architecture is typical for SAR-DFRC systems, with the DFRC waveform design module and demodulation module replaced by other methods.

III. Bistatic SAR Integrated Waveform Formulation and Solution

In this section, the matrix form of STFT is formulated at first. Then, the optimization problem of STFT spectrum-shaped waveform design for SAR-DFRC is formulated.

A. Problem Formulation

Let $\mathbf{x} = [x(1), x(2), \ldots, x(N)]^T$, and $\mathbf{x} \in \mathbb{C}^{N\times 1}$ denotes the discrete form of $x(t)$ with N the number of coded subpulses.

To embed the information of transmitter, e.g., track, beam, and altitude, into the waveform, consider the communication constraints, which null the specific (m,t)th grid of STFT

$$\left|\Psi_{m,t}(\mathbf{x})\right| \le b, \quad m \in \Omega_{t_c}, \quad t \in \Upsilon \quad (6)$$

where b is the upper bound of the (m,t)th grid of STFT. Ω_{t_c} represents a positive integer set consisting of frequency points, which are manually selected at the tth time. Υ is a positive integer set consisting of all the time grids divided by the short-time window with its maximum range N. In other words, information of the transmitter is embedded into the position of null frequency grids in STFT domain of the dual-function waveform.

The matrix form of STFT can be formulated as follows:

$$\Psi(\mathbf{x}) = \mathbf{F}^H \mathbf{Diag}(\mathbf{x})\mathbf{S} \quad (7)$$

where $\mathbf{F}$ is the $N \times N$ Fourier transform matrix, and $\mathbf{S} \in \mathbb{R}^{N\times N_T}$ consists of the shifted window function. Here, we take the rectangular window and set the overlap length of STFT to $L-1$ as a typical example. In other words, the number of nonzero elements in $\mathbf{s}_t, t = 1, \ldots, N_T$ is L. Therefore, $\mathbf{S} \in \mathbb{R}^{N\times(N-L+1)}$ can be formulated as follows:

$$\mathbf{S} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \vdots & 1 & \ddots & \vdots \\ 1 & \vdots & \ddots & 0 \\ 0 & 1 & & 1 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 \end{pmatrix}. \quad (8)$$

$\Psi_{m,t}(\mathbf{x})$ can be further formulated as follows:

$$\Psi_{m,t}(\mathbf{x}) = \mathbf{f}_m^H \tilde{\mathbf{X}} \mathbf{s}_t = \mathbf{f}_m^H \tilde{\mathbf{S}}_t \mathbf{x} \quad (9)$$

where $\tilde{\mathbf{X}} = \mathbf{Diag}(\mathbf{x})$, $\mathbf{f}_m$ is the mth column of the Fourier transform matrix $\mathbf{F}$, and $\mathbf{s}_t = [s_t(1), \ldots, s_t(N)]^T \in \mathbb{R}^{N\times 1}$ is the tth column of $\mathbf{S}$

$$\tilde{\mathbf{S}}_t = \mathbf{Diag}(\mathbf{s}_t) = \begin{pmatrix} s_t(1) & & 0 \\ & \ddots & \\ 0 & & s_t(N) \end{pmatrix}. \quad (10)$$

For bistatic SAR scenarios, gaps between the null and un-null grids may not be large enough, which can lead to demodulation performance degradation. Thus, the un-null passband frequency grids also have to be constrained. Besides, it is necessary to preventing the energy leakage out of bandwidth in frequency domain. According to [36], the frequency spectrum of $\mathbf{x}$ can be synthesized by the STFT of $\mathbf{x}$ under certain conditions, which means the traditional spectral constraint [37], [38], [39] can be realized through constraints in the time-frequency domain. Thus, we rewrite constrains of the STFT domain as follows:

$$\begin{aligned} &\left|\mathbf{f}_{m_1}^H \tilde{\mathbf{S}}_{t_1} \mathbf{x}\right| \le u, \quad m_1 \in \Omega_{t_s}, \quad t_1 \in \Upsilon \\ &\left|\mathbf{f}_{m_2}^H \tilde{\mathbf{S}}_{t_2} \mathbf{x}\right| \le b, \quad m_2 \in \Omega_{t_c}, \quad t_2 \in \Upsilon \\ &\left|\mathbf{f}_{m_3}^H \tilde{\mathbf{S}}_{t_3} \mathbf{x}\right| \ge \upsilon, \quad m_3 \in \Omega_{t_p}, \quad t_3 \in \Upsilon \end{aligned} \quad (11)$$

where u is the upper bound of the stopband grids of the STFT spectrum. υ is the lower bound of the unconstrained grids of the STFT spectrum. Ω_{t_s} stands for the set of stopband frequency points. Ω_{t_c} stands for the set of null frequency points embedding the communication information. Ω_{t_p} stands for the set of the unconstrained frequency points.

Without loss of generality, assume Ω_{t_s}, Ω_{t_c}, and Ω_{t_p} satisfy

$$\begin{aligned} &\Omega_{t_s} \cup \Omega_{t_c} \cup \Omega_{t_p} = \Omega_t \\ &\Omega_{t_s} \cap \Omega_{t_c} = \emptyset, \quad \Omega_{t_s} \cap \Omega_{t_p} = \emptyset, \quad \Omega_{t_c} \cap \Omega_{t_p} = \emptyset \end{aligned} \quad (12)$$

where Ω_t stands for the set of all the frequency points of the Fourier transform of $\tilde{\mathbf{S}}_t \mathbf{x}$.

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Again, to avoid the degradation or distortion while generating the waveform in the SAR system, consider the constant modulus constraint [40], [41]

$$|x(n)| = 1, \quad n = 1, 2, \ldots, N. \tag{13}$$

To prevent the sidelobe of strong scattering objects from masking the nearby weak objects in the range direction of the SAR image, WPSL [42] is introduced as the objective function in the following derivation and simplification.

Thus, the objective function can be represented as follows:

$$\sum_{n=1}^{N-1} \omega(n)|r(n)|^p \tag{14}$$

where the values of $\{\omega(n)\}_{n=1}^{N-1}$ are the nonnegative weights, $r(n)$ is the aperiodic autocorrelation of $\mathbf{x}$ at lag n, and $2 \leq p \leq \infty$.

Combining (11), (13), and (14) and substituting $\mathbf{f}_m^H \tilde{\mathbf{S}}_t$ with $\mathbf{q}_k^H$, $k = 1, \ldots, N \times N_T$, the dual-use waveform design problem can be posed as follows:

$$\begin{aligned} \min_{\mathbf{x}} \ & \sum_{n=1}^{N-1} \omega(n)|r(n)|^p \\ \text{s.t. } & \left|\mathbf{q}_{k_1}^H \mathbf{x}\right| \leq u, \quad k_1 \in K_s \\ & \left|\mathbf{q}_{k_2}^H \mathbf{x}\right| \leq b, \quad k_2 \in K_c \\ & \left|\mathbf{q}_{k_3}^H \mathbf{x}\right| \geq \upsilon, \quad k_3 \in K_p \\ & |x(n)| = 1, \quad n = 1, 2, \ldots, N \end{aligned} \tag{15}$$

where K_s is a set of positive integers. It is the set of subscripts reconstituted mapping the vector $\mathbf{f}_{m_1}^H \tilde{\mathbf{S}}_{t_1}, m_1 \in \Omega_{t_s}, t_1 \in \Upsilon$ to the vector $\mathbf{q}_{k_1}$. K_c and K_p are composed in the same way.

Problem (15) is an NP-hard optimization problem with nonconvex constraint.

B. Algorithm for the Proposed Optimization Problem

In this section, we propose the waveform design approach that attempts to realize dual function for the proposed SAR-DFRC system leveraging the ADMM framework, which is effective for dealing with separable objective functions [39].

Introduce three auxiliary variables

$$z_s = \mathbf{q}_{k_1}^H \mathbf{x}, \quad k_1 \in K_s \tag{16}$$

$$z_c = \mathbf{q}_{k_2}^H \mathbf{x}, \quad k_2 \in K_c \tag{17}$$

$$z_p = \mathbf{q}_{k_3}^H \mathbf{x}, \quad k_3 \in K_p. \tag{18}$$

Thus, problem (15) becomes

$$\begin{aligned} \min_{\mathbf{x}, z_s, z_s, z_p} \ & \sum_{n=1}^{N-1} \omega(n)|r(n)|^p \\ \text{s.t. } & z_s = \mathbf{q}_{k_1}^H \mathbf{x}, \quad k_1 \in K_s \\ & |z_s| \leq u, \quad s \in M^{K_s} \\ & z_c = \mathbf{q}_{k_2}^H \mathbf{x}, \quad k_2 \in K_c \\ & |z_c| \leq b, \quad c \in M^{K_c} \\ & z_p = \mathbf{q}_{k_3}^H \mathbf{x}, \quad k_3 \in K_p \\ & \left|z_p\right| \geq \upsilon, \quad p \in M^{K_p} \\ & |x(n)| = 1, \quad n = 1, 2, \ldots, N \end{aligned} \tag{19}$$

Algorithm 1 MMPSL-TFSS

Input $\mathbf{S}$, $\mathbf{x}^{(0)}$, ω, p, ρ, u, c, υ, Ω_{t_s}, Ω_{t_c}, Ω_{t_p}, $i = 0$

repeat

1: Update $\mathbf{x}^{(i+1)}$ by solving

$$\begin{aligned} \min_{\mathbf{x}} \ & \mathcal{L}_\rho\left(\mathbf{x}, z_s^{(i)}, z_c^{(i)}, z_p^{(i)}, \lambda_{1,s}^{(i)}, \lambda_{2,c}^{(i)}, \lambda_{3,p}^{(i)}\right) \\ \text{s.t. } & |x(n)| = 1, \quad n = 1, 2, \ldots, N. \end{aligned} \tag{20}$$

2: Update $z_s^{(i+1)}, z_c^{(i+1)}, z_p^{(i+1)}$ by solving

$$\begin{aligned} \min_{z_s, z_c, z_p} \ & \mathcal{L}_\rho\left(\mathbf{x}^{(i+1)}, z_s, z_c, z_p, \lambda_{1,s}^{(i)}, \lambda_{2,c}^{(i)}, \lambda_{3,p}^{(i)}\right) \\ \text{s.t. } & |z_s| \leq u, \quad s \in M^{K_s} \\ & |z_c| \leq b, \quad c \in M^{K_c} \\ & \left|z_p\right| \geq \upsilon, \quad p \in M^{K_p}. \end{aligned} \tag{21}$$

3: Update $\lambda_{1,s}^{(i+1)}, \lambda_{2,c}^{(i+1)}, \lambda_{3,p}^{(i+1)}$ by

$$\lambda_{1,s}^{(i+1)} = \lambda_{1,s}^{(i)} + \rho\left(z_s^{(i+1)} - \mathbf{y}_s^H \mathbf{x}^{(i+1)}\right), \quad s \in M^{K_s} \tag{22}$$

$$\lambda_{2,c}^{(i+1)} = \lambda_{2,c}^{(i)} + \rho\left(z_c^{(i+1)} - \mathbf{y}_c^H \mathbf{x}^{(i+1)}\right), \quad c \in M^{K_c} \tag{23}$$

$$\lambda_{3,p}^{(i+1)} = \lambda_{3,p}^{(i)} + \rho\left(z_p^{(i+1)} - \mathbf{y}_p^H \mathbf{x}^{(i+1)}\right), \quad p \in M^{K_p}. \tag{24}$$

4: $i = i + 1$;

until termination condition is satisfied

Output: Converged $\mathbf{x}^{(i+1)}$

where

$$\begin{aligned} M^{K_s} &= \{1, 2, \ldots, \text{Card}\{K_s\}\} \\ M^{K_c} &= \{1, 2, \ldots, \text{Card}\{K_c\}\} \\ M^{K_p} &= \left\{1, 2, \ldots, \text{Card}\left\{K_p\right\}\right\}. \end{aligned}$$

The solution of problem (19) via ADMM framework is summarized in Algorithm 1. This algorithm determines $\mathbf{x}, z_s, z_c, z_p, \lambda_{1,s}, \lambda_{2,c}$, and $\lambda_{3,p}$ in an alternating iterative fashion to minimize $\mathcal{L}_\rho(\mathbf{x}, z_s, z_c, z_p, \lambda_{1,s}, \lambda_{2,c}, \lambda_{3,p})$ by accounting for the constraints in problem (19). In particular, we denote by $\mathbf{x}^{(i)}, z_s^{(i)}, z_c^{(i)}, z_p^{(i)}, \lambda_{1,s}^{(i)}, \lambda_{2,c}^{(i)}$, and $\lambda_{3,p}^{(i)}$ the estimates of $\mathbf{x}, z_s, z_c, z_p, \lambda_{1,s}, \lambda_{2,c}$, and $\lambda_{3,p}$ from the ith iteration and report the ADMM procedure in Algorithm 1. The convergence proof in [39] still holds true to Algorithm 1 based on the same ADMM framework.

We call this algorithm the majorization–minimization PSL-TFSS (MMPSL-TFSS).

The augmented Lagrangian of problem (19) is given by

$$\begin{aligned} &\mathcal{L}_\rho\left(\mathbf{x}, z_s, z_c, z_p, \lambda_{1,s}, \lambda_{2,c}, \lambda_{3,p}\right) \\ &= \sum_{n=1}^{N-1} \omega(n)|r(n)|^p \\ &\quad + \sum_{\substack{s \in M^{K_s} \\ k_1 \in K_s}} \Re\left\{\lambda_{1,s}^*\left(z_s - \mathbf{q}_{k_1}^H \mathbf{x}\right)\right\} + \frac{\rho}{2} \sum_{\substack{s \in M^{K_s} \\ k_1 \in K_s}} \left\|z_s - \mathbf{q}_{k_1}^H \mathbf{x}\right\|^2 \end{aligned}$$

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$$+\sum_{\substack{c\in M^{K_c}\\ k_2\in K_c}}\Re\left\{\lambda_{2,c}^{*}\left(z_c-\mathbf{q}_{k_2}^{H}\mathbf{x}\right)\right\}+\frac{\rho}{2}\sum_{\substack{c\in M^{K_c}\\ k_2\in K_c}}\left\|z_c-\mathbf{q}_{k_2}^{H}\mathbf{x}\right\|^2 +\sum_{\substack{p\in M^{K_p}\\ k_3\in K_p}}\Re\left\{\lambda_{3,p}^{*}\left(z_p-\mathbf{q}_{k_3}^{H}\mathbf{x}\right)\right\}+\frac{\rho}{2}\sum_{\substack{p\in M^{K_p}\\ k_3\in K_p}}\left\|z_p-\mathbf{q}_{k_3}^{H}\mathbf{x}\right\|^2 \tag{25}$$

where $\rho>0$ is the penalty parameter, and $\lambda_{1,s}$, $\lambda_{2,c}$, and $\lambda_{3,p}$ are Lagrangian multipliers, respectively [43].

Next, we derive the solutions to problems (20) and (21).

1) Update $\mathbf{x}^{(i+1)}$ With $z_s^{(i)}, z_c^{(i)}, z_p^{(i)}, \lambda_{1,s}^{(i)}, \lambda_{2,c}^{(i)}$, and $\lambda_{3,p}^{(i)}$: Applying the majorization–minimization method [42], (14) can be further recast as follows:

$$\|\mathbf{x}-\mathbf{y}\| \tag{26}$$

where $\mathbf{y}$ is obtained via [42].

Thus, similar to [39], the objective function of problem (20) can be equivalently recast as follows:

$$\mathbf{x}^H\mathbf{R}\mathbf{x}+\Re\left\{\mathbf{d}^H\mathbf{x}\right\} \tag{27}$$

where

$$\mathbf{R}=\frac{\rho}{2}\left(\sum_{k_1\in K_s}\mathbf{q}_{k_1}\mathbf{q}_{k_1}^H+\sum_{k_2\in K_c}\mathbf{q}_{k_2}\mathbf{q}_{k_2}^H+\sum_{k_3\in K_p}\mathbf{q}_{k_3}\mathbf{q}_{k_3}^H\right) \tag{28}$$

$$\mathbf{d}=\Re\left\{\left[-2\mathbf{y}-\sum_{\substack{s\in M^{K_s}\\ k_1\in K_s}}\left(\lambda_{1,s}^{(i)}\mathbf{q}_{k_1}+\rho z_s^{(i)}\mathbf{q}_{k_1}\right)\right]\right\} +\Re\left\{\left[-\sum_{\substack{c\in M^{K_c}\\ k_2\in K_c}}\left(\lambda_{2,c}^{(i)}\mathbf{q}_{k_2}+\rho z_c^{(i)}\mathbf{q}_{k_2}\right)\right]\right\} +\Re\left\{\left[-\sum_{\substack{p\in M^{K_p}\\ k_3\in K_p}}\left(\lambda_{3,p}^{(i)}\mathbf{q}_{k_3}+\rho z_p^{(i)}\mathbf{q}_{k_3}\right)\right]\right\}. \tag{29}$$

Substituting $\mathbf{q}$ in (28) for $\tilde{\mathbf{S}}_t\mathbf{f}_m$, (28) becomes

$$\mathbf{R}=\frac{\rho}{2}\left(\sum_t\tilde{\mathbf{S}}_t\mathbf{J}\tilde{\mathbf{S}}_t\right) \tag{30}$$

where

$$\mathbf{J}=\left(\sum_{m_1\in\Omega_{t_s}}\mathbf{f}_{m_1}\mathbf{f}_{m_1}^H+\sum_{m_2\in\Omega_{t_c}}\mathbf{f}_{m_2}\mathbf{f}_{m_2}^H+\sum_{m_3\in\Omega_{t_p}}\mathbf{f}_{m_3}\mathbf{f}_{m_3}^H\right). \tag{31}$$

Due to the fact that $\Omega_{t_s}\cup\Omega_{t_c}\cup\Omega_{t_p}=\Omega_t$, $\mathbf{J}$ can be further simplified as $\mathbf{J}=N\mathbf{I}$. Therefore, $\mathbf{R}$ is a constant that is independent of the variable $\mathbf{x}$.

Thus, problem (20) can be further simplified as follows:

$$\begin{aligned}&\min_x\ \Re\left\{\mathbf{d}^H\mathbf{x}\right\}\\ &\text{s.t. } |x(n)|=1,\quad n=1,2,\ldots,N\end{aligned} \tag{32}$$

which can be solved with [44]

$$\mathbf{x}=e^{j\arg(-\mathbf{d})}. \tag{33}$$

$\mathbf{x}^{(i+1)}$ can be obtained by sequentially updating $\mathbf{y}$ and $\mathbf{x}$.

2) Update $z_s^{(i+1)}$, $z_c^{(i+1)}$, and $z_p^{(i+1)}$ With $\mathbf{x}^{(i+1)}$, $\lambda_{1,s}^{(i)}$, $\lambda_{2,c}^{(i)}$, and $\lambda_{3,p}^{(i)}$: Define

$$\tilde{z}_s^{(i+1)}=\mathbf{q}_{k_1}^H\mathbf{x}^{(i+1)}-\frac{1}{\rho}\lambda_{1,s}^{(i)},\quad k_1\in K_s \tag{34}$$

$$\tilde{z}_c^{(i+1)}=\mathbf{q}_{k_2}^H\mathbf{x}^{(i+1)}-\frac{1}{\rho}\lambda_{2,c}^{(i)},\quad k_2\in K_c \tag{35}$$

$$\tilde{z}_p^{(i+1)}=\mathbf{q}_{k_3}^H\mathbf{x}^{(i+1)}-\frac{1}{\rho}\lambda_{3,p}^{(i)},\quad k_3\in K_p. \tag{36}$$

Using the complete square method, the optimization problem (21) becomes

$$\begin{aligned}&\min_{z_s,z_c,z_p}\ f\left(z_s,z_c,z_p\right)\\ &\text{s.t. } |z_s|\le u,\quad s=1,\ldots,\text{Card}\{K_s\}\\ &\qquad |z_c|\le b,\quad c=1,\ldots,\text{Card}\{K_c\}\\ &\qquad \left|z_p\right|\ge v,\quad p=1,\ldots,\text{Card}\left\{K_p\right\}\end{aligned} \tag{37}$$

where

$$f\left(z_s,z_c,z_p\right)=\sum_s\frac{\rho}{2}\left|z_s-\tilde{z}_s^{(i+1)}\right|^2+\sum_c\frac{\rho}{2}\left|z_c-\tilde{z}_c^{(i+1)}\right|^2+\sum_p\frac{\rho}{2}\left|z_p-\tilde{z}_p^{(i+1)}\right|^2. \tag{38}$$

Thus, the solutions to problem (37) are

$$z_s^{(i+1)}=\begin{cases}u\dfrac{\tilde{z}_s^{(i+1)}}{\left|\tilde{z}_s^{(i+1)}\right|}, & \left|\tilde{z}_s^{(i+1)}\right|>u\\ \tilde{z}_s^{(i+1)}, & \text{otherwise}\end{cases} \tag{39}$$

$$z_c^{(i+1)}=\begin{cases}c\dfrac{\tilde{z}_c^{(i+1)}}{\left|\tilde{z}_c^{(i+1)}\right|}, & \left|\tilde{z}_c^{(i+1)}\right|>b\\ \tilde{z}_c^{(i+1)}, & \text{otherwise}\end{cases} \tag{40}$$

$$z_p^{(i+1)}=\begin{cases}v\dfrac{\tilde{z}_p^{(i+1)}}{\left|\tilde{z}_p^{(i+1)}\right|}, & \left|\tilde{z}_p^{(i+1)}\right|<v\\ \tilde{z}_p^{(i+1)}, & \text{otherwise.}\end{cases} \tag{41}$$

After obtaining the converged $\mathbf{x}^{(i+1)}$, $z_s^{(i+1)}$, $z_c^{(i+1)}$, and $z_p^{(i+1)}$, lagrangian multipliers $\lambda_{1,s}^{(i+1)}$, $\lambda_{2,c}^{(i+1)}$, and $\lambda_{3,p}^{(i+1)}$ can be calculated by (22)–(24), respectively.

Then, the converged dual-function waveform realizing information sharing for the bistatic SAR system is obtained when the termination condition is satisfied.

IV. Experimental and Numerical Results

This section addresses the performance assessment of the dual-use waveform in terms of the ACF performance, engineering realizability, reliability of communication, and the bistatic SAR imaging performance. The parameters of the waveform are shown in Table II with the penalty parameters $\rho=0.001$ and $p=10$.

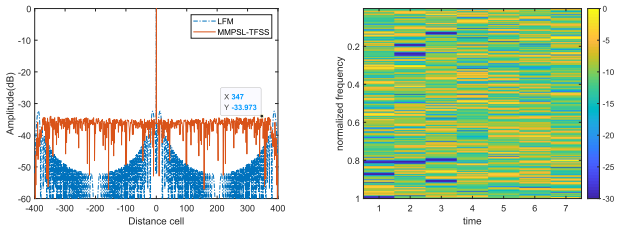
A. Performance of the Proposed Algorithm

This section aims at analyzing the ACF performance under different sidelobe weights. Set the stopband constraint

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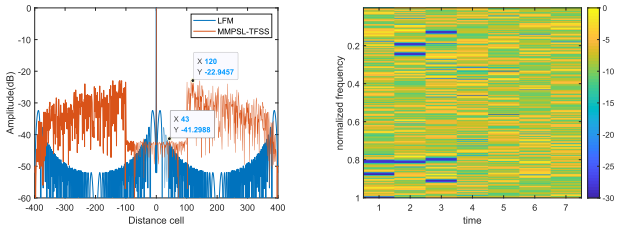
TABLE II
PARAMETERS OF WAVEFORM

Parameter	Value	Parameter	Value
Sampling rate	400 MHz	STFT window function	Rectangular
Sampling number	400	STFT window length	100
u, b, v	$3, 1, 6$	STFT overlap length	50



(a) (b)

Fig. 4. PSL waveform without stopband constraints. (a) ACF. (b) STFT spectrum.



(a) (b)

Fig. 5. WPSL waveform without stopband constraints. (a) ACF. (b) STFT spectrum.

$\mathbf{\Omega}_{t_s} = \emptyset$, $t = 1, \ldots, 7$, and the sidelobe weights in Fig. 4 is $\omega(n) = 6, \forall n$. For Fig. 5, the weights are assigned by

$$\omega(n) = \begin{cases} 10, & n = 3, \ldots, 100 \\ 1, & \text{otherwise.} \end{cases} \tag{42}$$

Since the shape of the spectrum determines the ACF, significant challenges will appear in balancing PSL performance and communication capacity of the frequency-domain-only DFRC waveform [29]. With an additional dimension, the time-frequency spectrum brings us a possibility to settle this contradiction. The comparison of PSL performances of MMPSL-TFSS and normalized PSL (NPSL) [29] is considered with -33.97 and -28.92 dB, respectively. The 5-dB lower PSL reveals that our proposed algorithm outperforms the NPSL [29] method.

Although Fig. 4(a) has a 11-dB lower PSL than Fig. 5(a), the level of the un-null area in Fig. 4(b) is lower than Fig. 5(b) averagely, due to the larger objective function value. The aforementioned lower level may shorten the gap between the null frequencies $\mathbf{\Omega}_{t_c}$ and the un-null frequencies $\mathbf{\Omega}_{t_p}$, which will lead to a more noise-susceptible communication performance. As a result, a trade-off must be made between communication constraints and PSL performance. In order to safeguard the reliability of communication, weights shown in Fig. 5(b) are used to proceed with the following experiment, which can also prohibit high sidelobes from masking weak targets to ensure the SAR imaging performance.

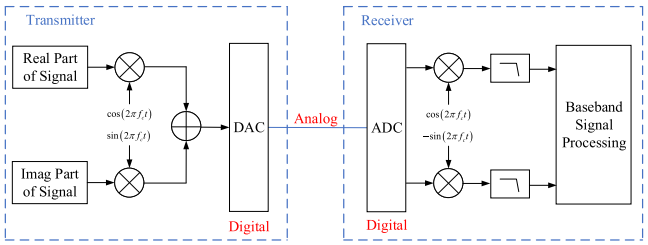


Fig. 6. Signal flow diagram of the system.

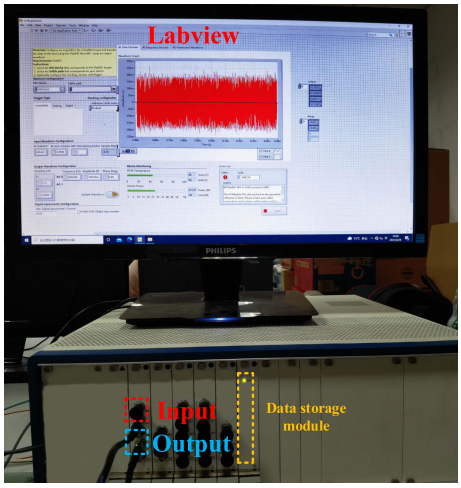


Fig. 7. Sketch of the semi-physical experiment platform.

B. Waveform Validation Experimental Results

In order to further assess the engineering realizability of the waveform, the semi-physical experiment implementation of SAR-DFRC waveforms to shape the STFT spectrum and ensure sidelobe performance is studied. More in detail, parameters are shown in Table II, and weights are shown in (42). The constraints are assigned by

$$\begin{aligned}
\mathbf{\Omega}_{t_s} &= [0.2525{:}0.0025{:}0.75], \quad t = 1, \ldots, 7 \\
\mathbf{\Omega}_{1_c} &= [0.8025{:}0.0025{:}0.8125] \\
&\quad \cup [0.865{:}0.0025{:}0.875] \cup [0.99{:}0.0025{:}1] \\
\mathbf{\Omega}_{2_c} &= [0.19{:}0.0025{:}0.2] \\
&\quad \cup [0.24{:}0.0025{:}0.25] \cup [0.8025{:}0.0025{:}0.8125] \\
\mathbf{\Omega}_{3_c} &= [0.1275{:}0.0025{:}0.1375] \\
&\quad \cup [0.79{:}0.0025{:}0.80] \cup [0.9025{:}0.0025{:}0.9125] \\
\mathbf{\Omega}_{t_c} &= \emptyset, \quad t = 4, \ldots, 7
\end{aligned}$$

where $\mathbf{\Omega}_{t_s}$ stands for the set of normalized stopband frequency points in the tth column of the STFT spectrum. $\mathbf{\Omega}_{t_c}$ stands for the set of normalized null frequency points. $\mathbf{\Omega}_{t_p}$ stands for the set of the unconstrained frequency points, obtained via (12).

The signal flow diagram of the experimental system is shown in Fig. 6. The waveforms are generated by via the semi-physical experiment platform shown in Fig. 7. The complex waveform is upsampled and in-phase/quadrature (I/Q) modulated with the carrier frequency $f_c = 1.2$ GHz. The analog signal is sampled and further digitally downconverted and recovered to the baseband complex signal. Since different

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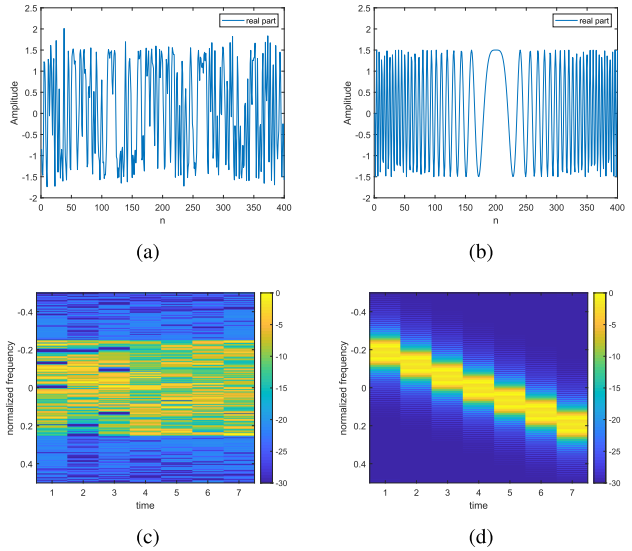


Fig. 8. Comparison of the waveform and LFM in the time and STFT domain. (a) Real part of the baseband waveform in the time domain. (b) Real part of LFM in the time domain. (c) STFT spectrum of the waveform. (d) STFT spectrum of LFM.

information corresponds to different nulling positions in the STFT domain, a specific circumstance is given as an example.

The real parts of the corresponding generated baseband waveform and LFM are shown in Fig. 8(a) and (b), respectively. Besides, the STFT spectrum of the corresponding waveform and LFM is shown in Fig. 8(c) and (d), respectively.

Unlike the linear energy distribution of LFM in the STFT spectrum shown in Fig. 8(d), our waveform controls the energy distribution over the entire STFT spectrum. This characteristic enables our waveform to exploit the high degree of freedom in the 2-D spectrum to carry information and optimize the PSL.

The modulation template, the STFT spectrum of the corresponding synthesized waveform, and the experimentally generated waveform are shown in Fig. 9(a)–(c), respectively. The amplitude of each communication cell in Fig. 9(b) is suppressed to at least -30 dB to mitigate the adverse effects of noise. The comparison of Fig. 9(a) and (b) indicates that our method can effectively control the energy in the time-frequency domain. Besides, the peak signal-to-noise ratio (PSNR) between Fig. 9(b) and (c) is 111.2 dB with 8 bits per pixel, which reveals the robustness of our waveform under the quantization error and the corresponding system noise.

The ACF of synthesized and experimentally sampled waveform is shown in Fig. 10. From Fig. 10(b), the first sidelobe level is suppressed to -21.9 dB compared with -13.4 dB of LFM, with the 3-dB mainlobe width almost the same. Actually, for SAR, the final ACF of range profile of a point target can be lower than Fig. 10 [45], which will be shown in Section IV-D. Combining Figs. 9 and 10, it verifies the engineering realizability of our waveform.

C. Demodulation Process and Performance

In this section, we discuss the communication performance analysis (capacity and BER) and confidentiality analysis of our waveform. The parameter M aforementioned in (2) is given by $M = N/2m = 40$, where $m = 5$.

TABLE III
COMMUNICATION CAPACITY COMPARISON

Algorithm	MMPSL-TFSS	NPSL	SBE-DSADMM
Communication capacity	93 bits	8 bits	9 bits

1) Communication Performance Analysis: According to (2), larger $\hat{L}$ brings larger communication capacity, and larger $\max\{\Gamma_c\}/\min\{\Gamma_p\}$ brings better BER performance [21], where Γ_c is the set consisting of null frequency grids in all columns of the time grids, and Γ_p consists of other frequency grids in all columns of the time grids. However, it is impossible to null all the spectra. As a result, $\hat{L}$ must be properly chosen.

Fig. 11 illustrates the impact of $\hat{L}$ affecting the value. The sequences with different $\hat{L}$ values are initialized with the Golomb sequence [46] after 200 iterations.

When $\hat{L} > 3$, the value rapidly declines. When $\hat{L} = 2$, the amount of information that can be embedded is not enough. Therefore, $\hat{L} = 3$ is chosen to perform the following results to meet the information sharing requirement. The maximum amount of information transmitted in one pulse is given by

$$E = \left\lfloor 7\log_2 \sum_{i=0}^{3} C_{40}^{i} \right\rfloor = 93 \text{ (bits)}. \tag{43}$$

The NPSL [29] algorithm embeds the information into the waveform by spectrum nulling. The sequential block enhancement—Dinkelbach's procedure, sequential convex approximation, and alternating direction method of multipliers (SBE-DSADMM) [31] algorithm designs the integrated waveform via spatiospectral modulation. The comparison of MMPSL-TFSS, NPSL, and SBE-DSADMM is carried out to verify the advantage of our method on communication capacity. Results are listed in Table III. The STFT spectrum is a 2-D spectrum containing time variant frequency information. As a result, a higher theoretical communication capacity can be achieved. Combing the PSL performance shown in Section IV-A, it highlights the effectiveness of the TFSS method in achieving low sidelobe level and large communication capacity compared with the frequency modulation DFRC waveforms.

A typical demodulation example is given as follows. First, the amplitude normalized passband frequency points are mapped to frequency grids with m frequency points in each grid, because the summation of the amplitude of the frequency points will smooth the noise. Therefore, the [0.0025:0.0025:0.0125] normalized frequency points are mapped to the first frequency grids. Again, [0.015:0.0025:0.25] normalized frequencies are mapped to the 2nd to 20th frequency grids, as shown in Fig. 12. Next, a threshold is applied to the mapped spectrum to demodulate the codes.

We reassign the whole plane as a binary plane

$$\Lambda_{m,t} = \begin{cases} 1, & \chi_{m,t} < \zeta \\ 0, & \chi_{m,t} \geq \zeta \end{cases} \tag{44}$$

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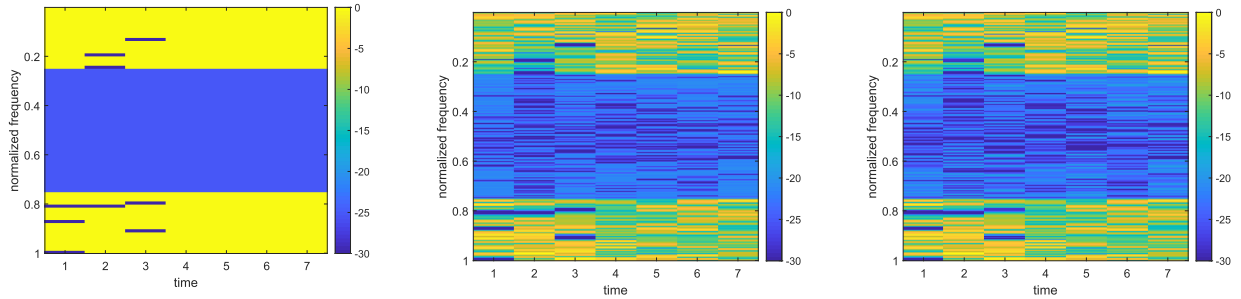


Fig. 9. Comparison of the STFT spectrum. (a) STFT spectrum modulation template. (b) STFT spectrum of the synthesized waveform. (c) STFT spectrum of the generated waveform.

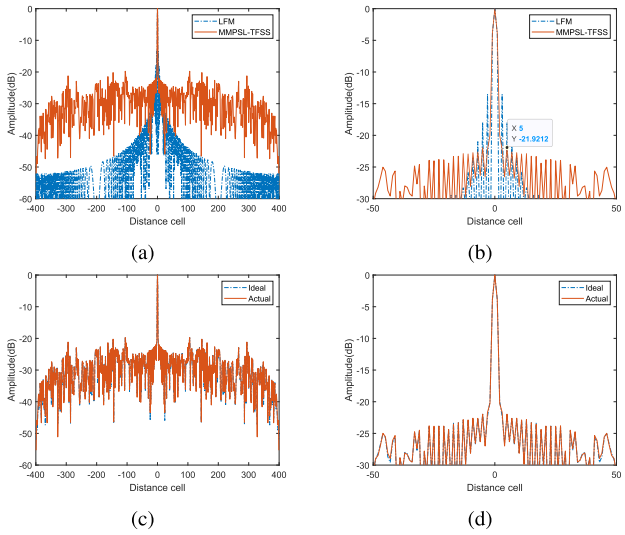


Fig. 10. Comparison of ACF. (a) ACF of the synthesized waveform versus LFM. (b) Enlarged ACF of the synthesized waveform versus LFM. (c) ACF of the synthesized waveform versus experimentally generated waveform. (d) Enlarged ACF of the synthesized waveform versus experimentally generated waveform.

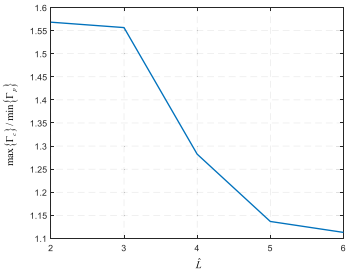


Fig. 11. $\max\{\Gamma_c\}/\min\{\Gamma_p\}$ versus $\hat{L}$ from 2 to 6.

where $\chi_{m,t} = \mathbf{f}_m^H \tilde{\mathbf{x}} \mathbf{s}_t$, and ζ is the threshold calculated by

$$\zeta = \left(\max_{\substack{m \in \Gamma_c \\ t \in \{1,2,3\}}} \{\chi_{m,t}\} + \min_{\substack{m \in \Gamma_p \\ t \in \{1,2,3\}}} \{\chi_{m,t}\} \right) \Big/ 2. \tag{45}$$

The BER versus SNR is calculated within the first three time grids and performed with 10 000 Monte Carlo experiments, as shown in Fig. 13. In particular, BER decreases to less than 1% with SNR larger than 18 dB.

2) Confidentiality Analysis: This section aims at analyzing the confidentiality of the proposed TFSS waveform. As mentioned before, the STFT requires the transform window length

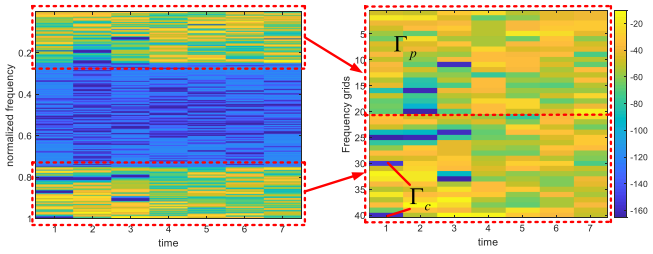


Fig. 12. Mapping of the frequency grids.

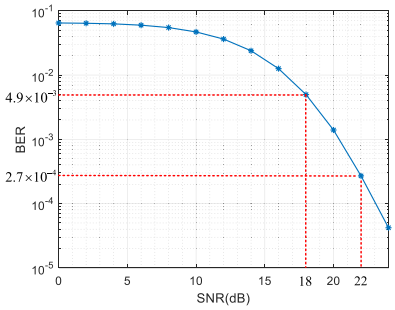


Fig. 13. BER versus SNR.

L and overlap length O to be known. In other words, information can only be correctly demodulated with the correct length and overlap, which double encrypts our waveform.

Fig. 14(a) is the correct demodulating result with a 40- $\times$ 7-bits code. Reassigning the value of the overlap length O to 40 and 80, the number of the code bits becomes 40 $\times$ 6 and 40 $\times$ 16, as shown in Fig. 14(c) and (e), respectively. Keeping O constant and changing the window length L to 150 and 80, the number of the code bits now becomes 40 $\times$ 3 and 40 $\times$ 11, as shown in Fig. 14(b) and (d), respectively.

Based on the results above, we can conclude that both L and O can affect the STFT spectrum. The effect of L is more obvious than that of O. The two parameters can impact the amplitudes, positions of null grids, and, more importantly, the number of code bits, making it even more difficult for noncooperative receivers to decipher than most of the existing DFRC waveforms.

D. Bistatic SAR Imaging Results

In this section, we perform spotlight bistatic SAR simulation with waveforms experimentally generated in Section IV-B

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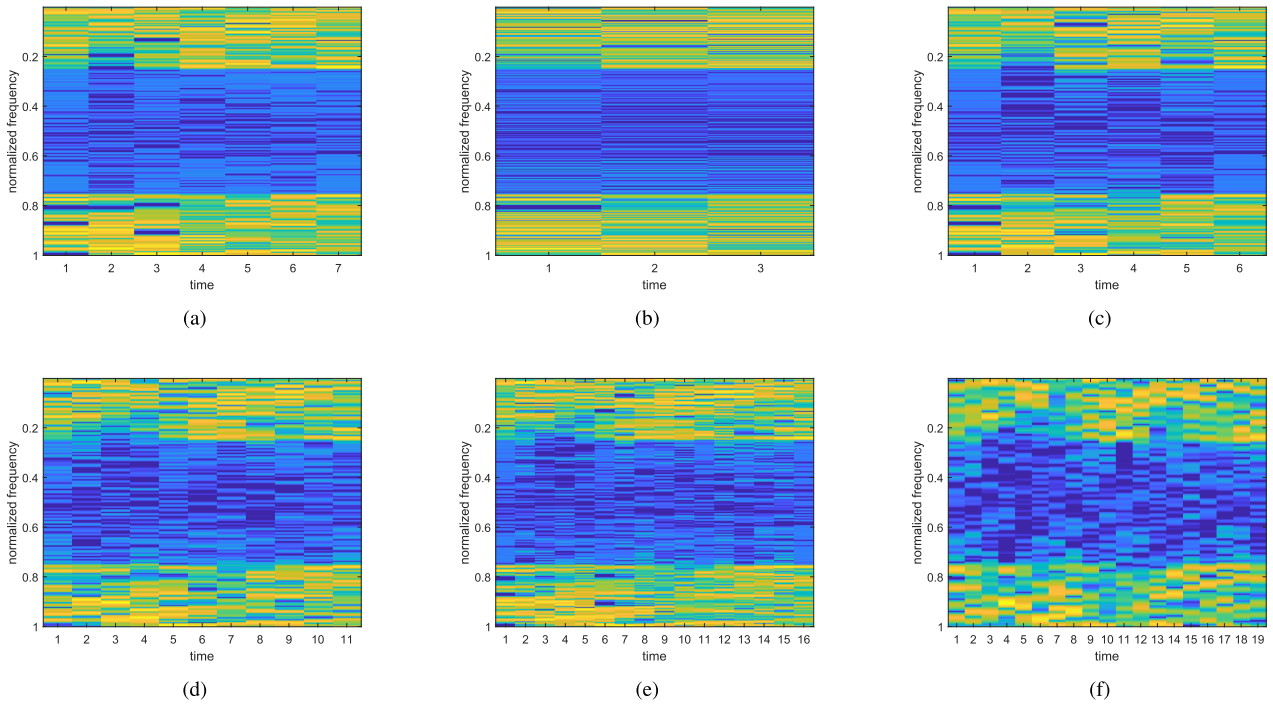


Fig. 14. Different parameters applied to STFT. (a) $L = 100$, $O = 50$. (b) $L = 150$, $O = 50$. (c) $L = 100$, $O = 40$. (d) $L = 80$, $O = 50$. (e) $L = 100$, $O = 80$. (f) $L = 40$, $O = 20$.

TABLE IV
PARAMETERS OF SIMULATION

Parameter	Symbols name	Value
Carrier frequency	f_c	9.6 GHz
Pulse repetition frequency	f_r	1000 Hz
Range sampling number	N_r	4096
Azimuth sampling number	N_a	4000
Transmitter initial position	$[X_T, Y_T, H_T]$	$[-2000, -400, 2980]$ m
Receiver initial position	$[Y_T, Y_R, H_R]$	$[0, -400, 1500]$ m
Transmitter velocity	$[v_{Tx}, v_{Ty}, v_{Tz}]$	$[100, 200, 10]$ m/s
Receiver velocity	$[v_{Rx}, v_{Ry}, v_{Rz}]$	$[0, 200, 0]$ m/s

and scattering coefficients obtained by actual SAR image to verify the capability of communication and imaging. With the simulation parameters we set, the transmitter moves along a straight line, and the receiver moves in the positive direction of the y-axis, as shown in Fig. 1. Other parameters are shown in Table IV.

In our scenario, transmitter coordinates are embedded into the STFT spectrum as the transmitted information, as shown in Fig. 15, which is used for the receiver to perform the SAR imaging algorithm along with the echo. But, updating the coordinate in each pulse is not always necessary, because simple tracks can be further fit by finite coordinate points. Therefore, the same waveform is transmitted over several pulses. Here, we embed the transmitter coordinate under a 10-Hz refreshing rate into the time-frequency spectrum of phase coded signals. In other words, the waveform is updated by the frequency of 10 Hz with a total number of 40.

Notice that a synchronization bit can be embedded between the waveforms before and after coordinate refreshing, similar to PSK.

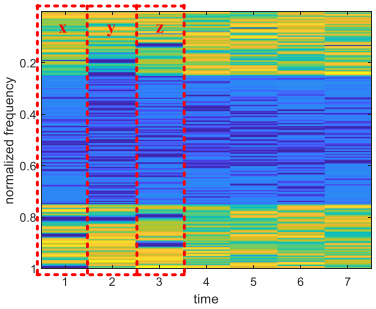


Fig. 15. Schematic of information embedded to transmitted waveform.

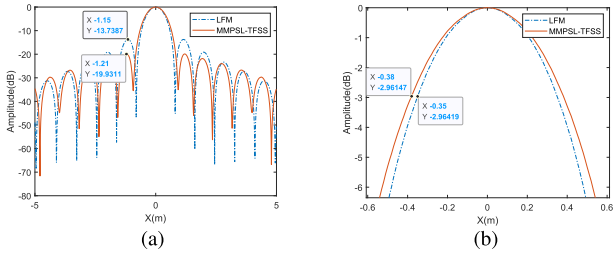


Fig. 16. Point target range profile. (a) Sidelobe analysis. (b) Mainlobe analysis.

An alphabet between the null areas in the STFT spectrum and space coordinates must be designed to code the information. We shall first quantify the space. According to Table IV, the space whose range is $x \in [-2300, -1299]$, $y \in [-500, 499]$, and $z \in [2960, 3039]$ is quantified by 10 000 points.

The x, y, and z coordinates are divided by 10 000 points, because we only concern about the communication capability.

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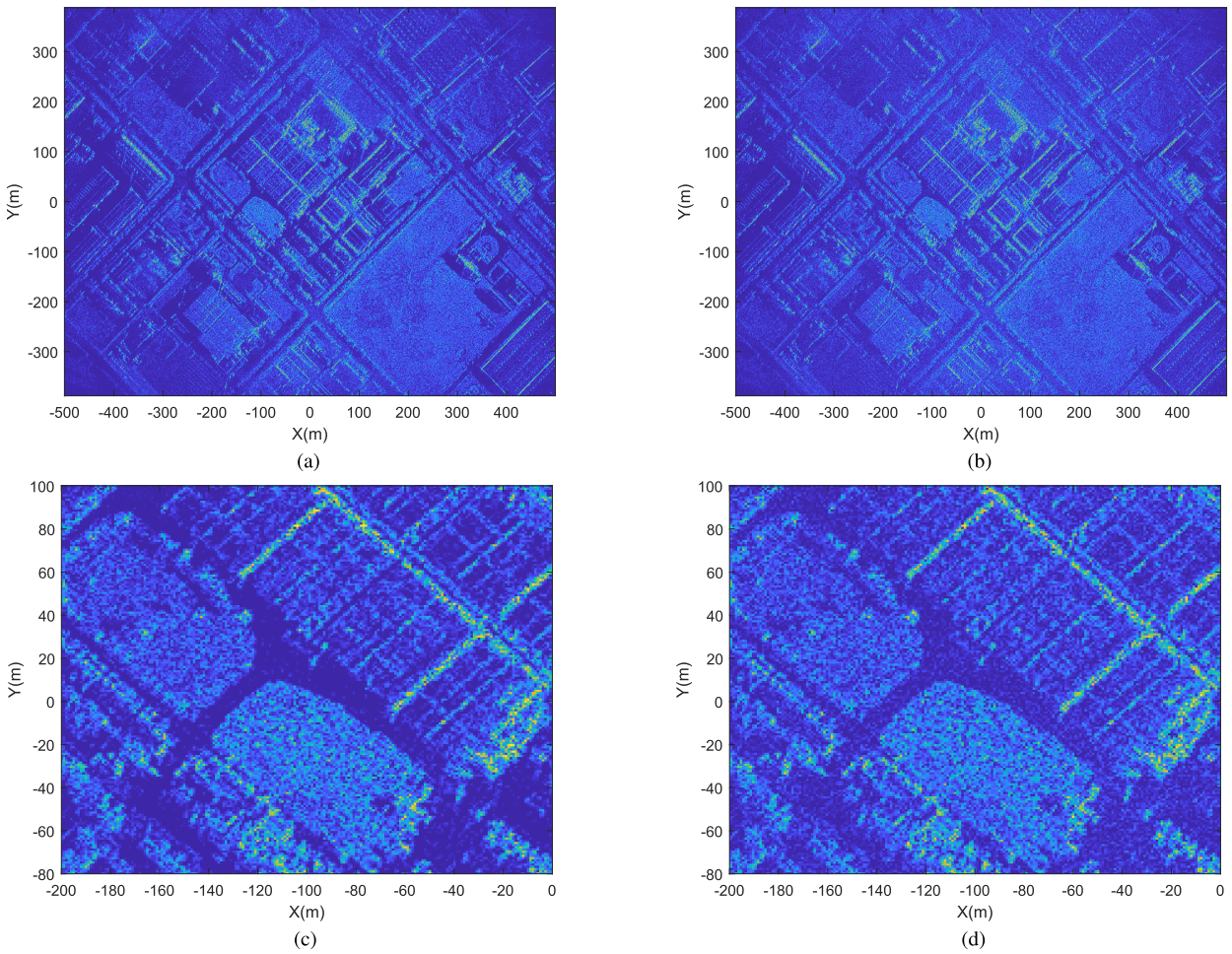


Fig. 17. Bistatic SAR imaging results. (a) LFM. (b) Sequences designed by the proposed method. (c) Enlarged image view of the LFM. (d) Enlarged image view of the proposed sequences.

As for the encoding and quantization methods, they will directly or indirectly affect the imaging quality indeed. But, they are irrelevant to the waveform performance that needs to be verified. Therefore, coordinates are quantified to be consistent with the original position as close as possible to reduce the impact on imaging performance due to the quantization error. Again, the positions of null frequency grids in the STFT spectrum corresponding to the y coordinate are flipped to avoid the particular case of the same null position at the first and second columns of the STFT spectrum.

Based on (43), the amount of statuses one time grid can cover is

$$E_t = \sum_{i=0}^{3} C_{40}^{i} = 10\,101. \tag{46}$$

We select the first 120 statuses of C_{40}^{2} and all statuses of C_{40}^{3} to form a total of 10 000 statuses. Thus, the $x/y/z$ coordinates of the transmitter are mapped to each of the 10 000 statuses of the designed STFT spectrum correspondingly. In other words, information is embedded as combinations of different positions of the null frequency grids.

For example, the initial x coordinate of the transmitter is -2000; then, it corresponds to the 3000th state in all 10 000 statuses, which corresponds to the [0.8025:0.025:0.8125], [0.865:0.025:0.875], and [0.99:0.025:1] null normalized frequency points.

As mentioned ahead, the coordinate refresh rate is 10 Hz, which does not correspond to 1000-Hz PRF of the echo. However, the backprojection (BP) algorithm requires both transmitter and receiver coordinates at each PRF. Therefore, curve fitting is carried out to increase the coordinate refresh rate to 1000 Hz. Then, 40 coordinate information carried by 40 waveforms is fit to 4000 coordinates to achieve the imaging requirement.

Fig. 16 shows the range profile of point target imaging results with LFM and our waveform. The result indicates that our proposed waveform can suppress the PSL to -19.9 dB with the 3-dB mainlobe widened around 9%.

In addition to point target simulation, a bistatic SAR scene simulation whose echo is generated from an actual SAR image with 1- $\times$ 1-m grids is also conducted. The result is shown in Fig. 17.

Image contrasts of Fig. 17(a) and (b) are 0.0615 and 0.0589, respectively, and the entropies are 12.4178 and 12.5928, respectively. The PSNR of the two images is 81.35 dB with 8 bits per pixel. These numerical results verify that our waveform can achieve an almost identical SAR imaging

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performance compared with LFM under the same bandwidth while carrying information.

V. Conclusion

This article proposes a novel SAR-DFRC architecture and a novel DFRC waveform design method. The novel DFRC waveform realizes line-of-sight (LOS) downlink broadcast communication by nulling in the prescribed time-frequency grids. Besides, the SAR imaging performance is ensured by suppressing WPSL of the ACF. The waveform is optimized via the ADMM framework. The transmitter embeds its $x/y/z$ coordinates under a 10-Hz fresh rate into the first three columns of time grids in the STFT spectrum of a phase coded signal. The received echo is processed parallel. The coordinates are decrypted with transform length and overlap length of STFT. Then, the coordinates are demodulated by threshold decision and status index, as shown in Sections IV-C and IV-D, respectively. Coordinates are fit to match the echo in each pulse for imaging processing. The semi-physical experiments are conducted to verify the effectiveness of our waveform and the resolved contradiction between the PSL performance and the communication capacity. Numerical results verify the communication and SAR imaging performances of our waveform.

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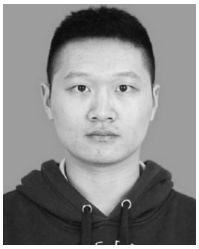
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Youshan Tan (Student Member, IEEE) received the B.S. degree in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2020, where he is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering.

His research interests include synthetic aperture radar (SAR) imaging and waveform design.



Zhongyu Li (Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2011 and 2017, respectively.

From 2015 to 2016, he was a Visiting Ph.D. Student with the Department of Information Engineering, Electronics and Telecommunications, Sapienza University of Rome, Rome, Italy. He is currently an Associate Professor with the UESTC. His research interests include synthetic aperture radar technology (particular emphasis on synthetic aperture radar ground moving target detection and imaging).

Dr. Li is a Reviewer of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and the *Journal of Electromagnetic Waves and Applications*.



Jing Yang received the B.S. and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2017 and 2022, respectively.

From 2019 to 2020, she was a Visiting Researcher with the University of Naples Federico II, Naples, Italy. She is currently a Lecturer with Zhengzhou University, Zhengzhou, China. Her research interests include array signal processing, integrated sensing and communication, and waveform optimization.



Xianxiang Yu received the B.S. and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2014 and 2020, respectively.

From 2018 to 2019, he was a Visiting Researcher with Pennsylvania State University, State College, PA, USA. He is currently a Post-Doctoral Fellow with UESTC. His research interests include multiple-input multiple-output (MIMO) radar, array signal processing, and waveform optimization.



Hongyang An (Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2015 and 2020, respectively.

From 2019 to 2020, he was a Visiting Student with Nanyang Technological University, Singapore. He is currently with the University of Electronic Science and Technology of China. His research interests include synthetic aperture radar and sparse signal processing.



Junjie Wu (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2004, 2007, and 2013, respectively, all in electronic engineering.

From 2012 to 2013, he was a Visiting Student with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA. He is currently a Professor with the UESTC. His research interests include synthetic aperture radar imaging (particular emphasis on bistatic synthetic aperture radar).

Dr. Wu is a Reviewer of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), TRANSACTIONS ON SIGNAL PROCESSING (TSP), Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS), Journal of Selected Topics in Signal Processing (JSTSP), Geoscience and Remote Sensing (GRS) Letters, and the IET Radar Sonar and Navigation (IET RSN).



Jianyu Yang (Member, IEEE) received the B.S. degree in electronic engineering from the National University of Defense Technology, Changsha, China, in 1984, and the M.S. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1987 and 1991, respectively.

He is currently a Professor with the School of Information and Communication Engineering, UESTC. His research interests include synthetic aperture radar (SAR) and statistical signal processing.

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